# EDS 223 Homework #3: Oyster & Dungeness Crab Suitability Analysis

**Geospatial Analysis & Remote Sensing**

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### Preparing Data

```
# Read in packages
library(tidyverse)
library(stars)
library(sf)
library(here)
library(tmap)
library(gt)
library(viridisLite)
```

```r
library(terra)
library(geodata)
library(kableExtra)
```

## Reading in data

```r
# Read in West Coast data
wc_EEZ <- sf::st_read(here::here('data/wc_regions_clean.shp'))
```

```
Reading layer `wc_regions_clean' from data source
  `/Users/swaggerboy/Documents/eds223/eds223-HW3/data/wc_regions_clean.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

```r
# Read in depth raster data
depth <- rast('data/depth.tif')

# Read in yearly annual sst tif files
rast_08 <- rast('data/average_annual_sst_2008.tif')
rast_09 <- rast('data/average_annual_sst_2009.tif')
rast_10 <- rast('data/average_annual_sst_2010.tif')
rast_11 <- rast('data/average_annual_sst_2011.tif')
rast_12 <- rast('data/average_annual_sst_2012.tif')
```

### Preparing Data

```r
# Verify CRS's match
if (st_crs(wc_EEZ) == st_crs(depth)){
    print('CRS is a match :)')
}else{
    warning('It is not a match:()')
}
```

```
[1] "CRS is a match :)"
```

```
# Combine all raster SST rasters
rasters <- c(rast_08, rast_09, rast_10, rast_11, rast_12)
```

### Process Data

```
# Calculate average
avg_sst <- mean(rasters, na.rm=TRUE)

# Converting average SST from Kelvin to Celsius
avg_sst <- avg_sst - 273.15

# Crop depth raster to match the extent of the SST raster
# depth_cropped <- terra::extend(depth, avg_sst)

# Resample depth to match avg_sst
depth_resampled <- terra::resample(x = depth, y = avg_sst, method = 'bilinear')

# Check extent, resolution, and CRS match for both data
if (st_crs(depth_resampled) == st_crs(avg_sst)){
    print('The first matches the second :)')
}else{
    warning('The first does not match the second >:()')
}
```

```
[1] "The first matches the second :)"
```

```
# CRS matches
# Extent matches
# Resolution matches

# CRS matches but EPSG codes do not
# Reprojecting depth_resampled
depth_resampled <- terra::project(depth_resampled, avg_sst)

# Stack matching CRS geodataframes
depth_sst <- c(depth_resampled, avg_sst)
```

### Find suitable locations

```
# Create reclassification matrix for sea surface temp using suitable sea surface temperatures
rcl_sst <- matrix(c(
    11, 30, 1,
    -Inf, 11, 0,
    31, Inf, 0),
    ncol = 3,
    byrow = TRUE
)

# Create reclasification matrix for depth using suitable conditions of depth for oysters
rcl_depth <- matrix(c(
    -70, 0, 1,
    -Inf, -70, 0,
    0, Inf, 0),
    ncol = 3,
    byrow = TRUE
)

# Reclassification of SST and depth
reclassified_sst <- classify(avg_sst, rcl = rcl_sst)
reclassified_depth <- classify(depth_resampled, rcl = rcl_depth)
```

### Finding locations that satisfy both SST and depth conditions

```
oyster_algebra <- lapp(c(reclassified_depth, reclassified_sst), function(a, b) a*b)
```

### Determine the most suitable EEZ

```
# Rasterize stacked raster for reclassified depth and SST
wc_raster <- as.factor(rasterize(wc_EEZ, oyster_algebra, field = 'rgn_id'))

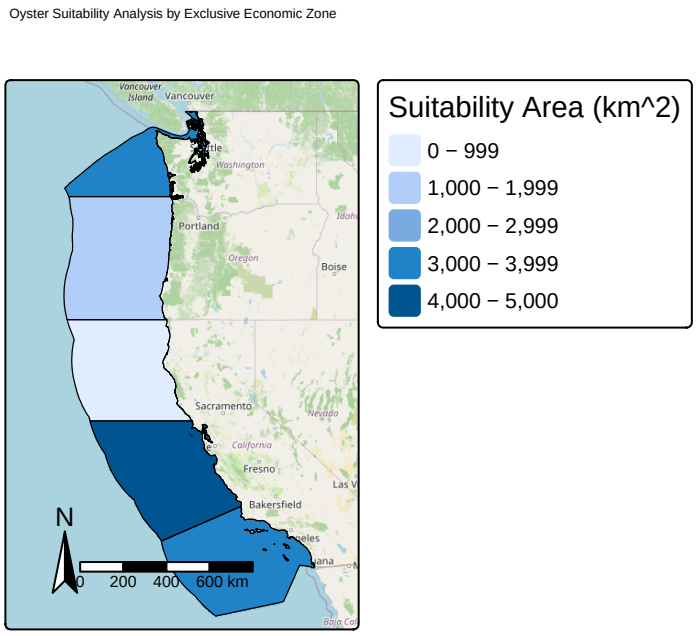
# Calculate cell area from wc_raster
area <- cellSize(wc_raster, unit ='km')

# Calculate total suitable area per zone
suitable_area <- lapp(c(area, oyster_algebra), function(a,b) a*b)
```

```
# Create suitablity areas for oyster
wc_suitability <- zonal(suitable_area, wc_raster, fun = sum, na.rm = TRUE, as.raster = TRUE)
```

### Mapping

```
# Create map for oyster suitability
tm_basemap('OpenStreetMap') +
    tm_shape(wc_suitability) +
        tm_raster(
            col.legend = tm_legend('Suitability Area (km^2)',
                                   position = tm_pos_out('right', 'center'))
    ) +
    tm_shape(wc_EEZ) +
        tm_borders(col = 'black',
                   lwd = 0.6,
                   lty = 'solid') +
    tm_title('Oyster Suitability Analysis by Exclusive Economic Zone') +
    tm_compass(position = c('left', 'bottom')) +
    tm_scalebar(position = c('center', 'bottom'))
```



The map produced suggests that the central/southern Californian coast is the most suitable for aquaculture operations based on the optimal growing conditions for oysters.

### Creating a function for most suitable Exclusive Economic Zones by species of interest

```
# Call function 'suitability_eez'
suitability_eez <- function(tmin = 11, tmax = 30, dmin = -70, dmax = 0, species = 'Oyster'){

# Create reclassification matrix for sea surface temp using suitable sea surface temperatures
rcl_sst <- matrix(c(
    tmin, tmax, 1,
    -Inf, tmin, 0,
    tmax, Inf, 0),
    ncol = 3,
    byrow = TRUE
)

# Create reclasification matrix for depth using suitable conditions of depth for oysters
rcl_depth <- matrix(c(
    dmin, dmax, 1,
    -Inf, dmin, 0,
    dmax, Inf, 0),
    ncol = 3,
    byrow = TRUE
)

# Reclassification of SST and depth
reclassified_sst <- classify(avg_sst, rcl = rcl_sst)
reclassified_depth <- classify(depth_resampled, rcl = rcl_depth)

# Verify that reclassified rasters' CRS match
if (st_crs(reclassified_sst) == st_crs(reclassified_depth)){
    print("CRS for both reclassified layers match.")
}else{
    stop("CRS does not match for both reclassified layers!")
}

# Multiply the raster cell values using lapp() to find locations that satisfy both SST and de
oyster_algebra <- lapp(c(reclassified_depth, reclassified_sst), function(a, b) a*b)

# Rasterize stacked raster for reclassified depth and SST
```

```
wc_raster <- as.factor(rasterize(wc_EEZ, oyster_algebra, field = 'rgn_id'))

# Calculate cell area from wc_raster
area <- cellSize(wc_raster, unit ='km')

# Calculate total suitable area per zone
suitable_area <- lapp(c(area, oyster_algebra), function(a,b) a*b)

# Create suitablity areas for oyster
wc_suitability <- zonal(suitable_area, wc_raster, fun = sum, na.rm = TRUE, as.raster = TRUE)
names(wc_suitability) <- 'Suitable Area (km^2)'

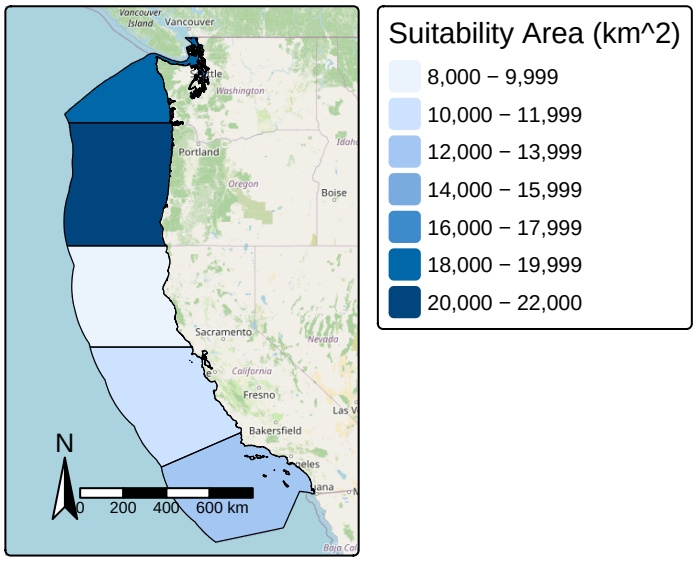
# Creating the map
tm_basemap('OpenStreetMap') +
    tm_shape(wc_suitability) +
        tm_raster(
            col.legend = tm_legend('Suitability Area (km^2)',
                                   position = tm_pos_out('right', 'center'))
    ) +
    tm_shape(wc_EEZ) +
        tm_borders(col = 'black',
                   lwd = 0.6,
                   lty = 'solid') +
    tm_title(paste0(species, ' Suitability Analysis by Exclusive Economic Zone')) +
    tm_compass(position = c('left', 'bottom')) +
    tm_scalebar(position = c('center', 'bottom'))
}
```

**Using the function for The Dungeness Crab (Metacarcinus magister)**

```
# Calling function for dungeness crab
suitability_eez(tmin = 3, tmax = 19, dmin = -360, dmax = 0, species = "Dungeness Crab")
```

```
[1] "CRS for both reclassified layers match."
```

Dungeness Crab Suitability Analysis by Exclusive Economic Zone




We can use the function to map the Dungeness Crab suitability analysis by Exclusive Economic Zone. The dungeness crab has a growth surface temperature range of 3 - 19 degrees Celsius and a depth range of 360 to 0 feet below sea level. The map suggests that the Exclusive Economic Zone on the coast of Oregon would be the most suitable.